

# Official-source Electricity compilation 15

Mixed-source chapter worksheet • cross-paper reuse disclosed

Total: 35 marks • Suggested time: 70 minutes

## Instructions and reuse disclosure

Curated compilation of official CBSE source questions, not a CBSE-issued chapter paper. Original source labels and exact source images are retained.

Answer every selected question. Only one selected source-slot alternative appears in this paper; do not answer or count excluded alternatives.

Questions recur across this 15-paper collection. Reuse is disclosed per question in the source appendix; these are not 15 disjoint official papers.

Sections use available source marks: A MCQ; B 2 marks; C 3 marks; D 5 marks or source-marked case questions. Case marks are not inflated to five.

## Section A — MCQ

Compilation questions 1-4; 4 marks. Exact source labels remain attached to each question.

## Section B — 2-mark questions

Compilation questions 5-8; 8 marks. Exact source labels remain attached to each question.

## Section C — 3-mark questions

Compilation questions 9-11; 9 marks. Exact source labels remain attached to each question.

## Section D — 5-mark / case-based questions

Compilation questions 12-14; 14 marks. Exact source labels remain attached to each question.

## Section A — MCQ

### 1. [1 marks] Source Q6 • 2019-20

- 6      When a 4V battery is connected across an unknown resistor there is a current of 100 mA in the circuit. The value of the resistance of the resistor is:
- i)  $4\ \Omega$
  - ii)  $40\ \Omega$
  - iii)  $400\ \Omega$
  - iv)  $0.4\ \Omega$

Source/editorial note: Original spelling “resister” retained. Q6 mark is printed on the preceding page (page 3); Section A instructions and the MS both confirm 1 mark.

## 2. [1 marks] Source Q13 • 2022-23

|    |                                                                                                                                                                                                                                                                                                       |   |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 13 | A complete circuit is left on for several minutes, causing the connecting copper wire to become hot. As the temperature of the wire increases, the electrical resistance of the wire<br>(a) decreases.<br>(b) remains the same.<br>(c) increases.<br>(d) increases for some time and then decreases . | 1 |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

## 3. [1 marks] Source Q14 • main

**14.** An electric iron of resistance  $20\ \Omega$  draws a current of  $5\text{ A}$ . The heat developed in the iron in 30 seconds is :

- (A) 15000 J (B) 6000 J  
(C) 1500 J (D) 3000 J

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

## 4. [1 marks] Source Q31 • 2026-27

|    |                                                                                                                                                                                              |   |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 31 | An electrical appliance rated $1000\text{ W}$ is operated at $250\text{ V}$ . What current does it draw?<br>A. $0.25\text{ A}$<br>B. $2.5\text{ A}$<br>C. $4.0\text{ A}$<br>D. $10\text{ A}$ | 1 |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

Source/editorial note: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

## Section B — 2-mark questions

### 5. [2 marks] Source Q27 • main

The values of current ( $I$ ) flowing through a given resistor of resistance ( $R$ ), for the corresponding values of potential difference ( $V$ ) across the resistor are as given below :

| V (volts)   | 0.5 | 1.0 | 1.5 | 2.0 | 2.5 | 3.0 | 4.0 | 5.0 |
|-------------|-----|-----|-----|-----|-----|-----|-----|-----|
| I (amperes) | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.8 | 1.0 |

Plot a graph between current ( $I$ ) and potential difference ( $V$ ) and determine the resistance ( $R$ ) of the resistor.

Source/editorial note: Occurrence mappings retained verbatim; only primary PDF/question evidence is accepted and counted. Other occurrences require their own evidence validation. Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

6. [2 marks] Source Q26 • 2017-18

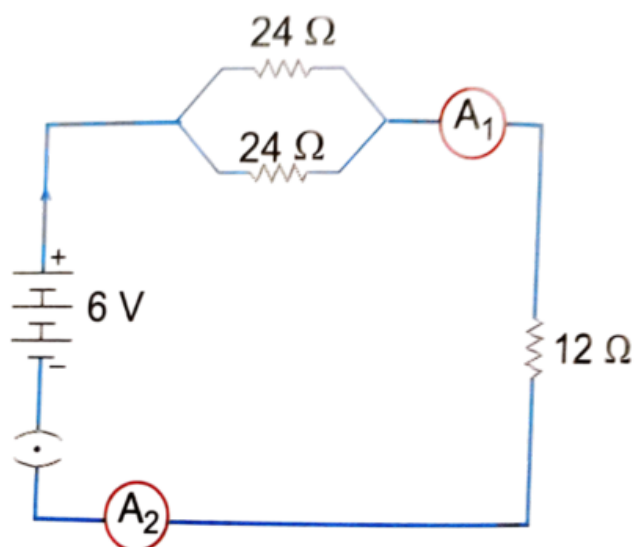
|              |                                                                                                                                                     |     |     |      |      |   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|------|------|---|
| 26           | The values of current $I$ flowing in a given resistor for the corresponding values of potential difference $V$ across the resistor are given below: |     |     |      |      | 2 |
|              |                                                                                                                                                     |     |     |      |      |   |
| $I$ (ampere) | 0.5                                                                                                                                                 | 1.0 | 2.0 | 3.0  | 4.0  |   |
| $V$ (volt)   | 1.6                                                                                                                                                 | 3.4 | 6.7 | 10.2 | 13.2 |   |

Plot a graph between  $V$  and  $I$  and calculate the resistance of the resistor.

Source/editorial note: Practical-skills question; continues from SQP page 3 to page 4. OR alternative stored separately. Official MS states  $3.2 \Omega$  and does not supply the plotted graph; data ratios are approximate (about  $3.2$ – $3.4 \Omega$ ). The official worked slope uses graph coordinates not listed in the table; preserved rather than silently corrected. Official answer unavailable/withheld for an identified error, omission or incomplete verification: Official slope solution captured, but required plotted graph absent from official MS; not a complete publishable model answer. Consult the linked original marking scheme with this limitation in mind.

7. [2 marks] Source Q34 B • 2025-26

B.



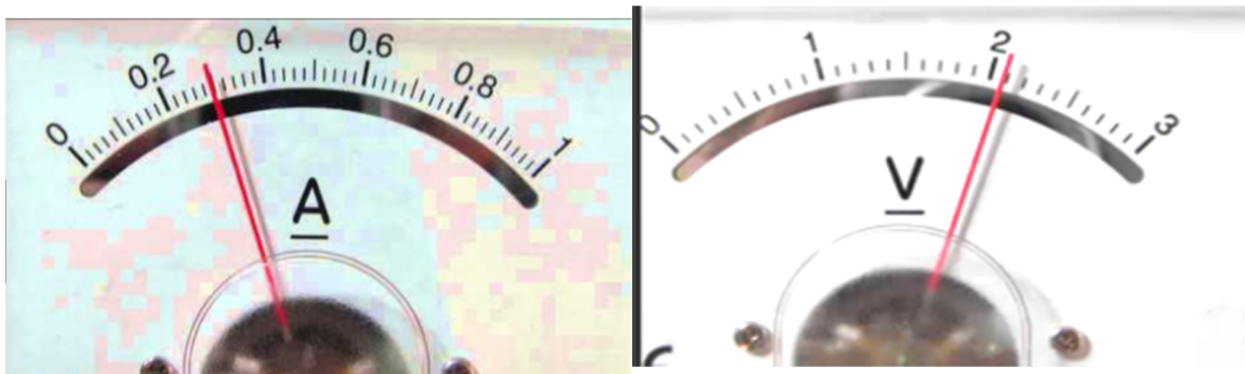
Study the circuit and find out-

- Current in  $12 \Omega$  resistor
- Difference in the readings of ammeter  $A_1$  and  $A_2$  if any

Source/editorial note: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

## 8. [2 marks] Source Q26 • 2018-19

26. The current flowing through a resistor connected in an electrical circuit and the potential difference developed across its ends are shown in the given ammeter and voltmeter. Find the least count of the voltmeter and ammeter. What is the voltage and the current across the given resistor? 2



Source/editorial note: Practical-skills question. Official SQP layout error: the ammeter/voltmeter image appears beside chemistry Q22 near the top of the SAME PDF page 5, not alongside Q26. The matching meters and MS establish the intended association. Evidence array puts Q26 text first and displaced diagram second. No image was redrawn.

## Section C — 3-mark questions

### 9. [3 marks] Source Q27 • main

- 27.** (a) Define one ampere.
- (b) The resistance of a wire of 0.01 cm radius is  $14 \, \Omega$ . If the resistivity of the material of the wire is  $44 \times 10^{-8} \, \Omega\text{m}$ , find the length of the wire. (Given  $\pi = \frac{22}{7}$ ) 3

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

### 10. [3 marks] Source Q36 • 2026 First Board



- 36.** The values of current  $I$  flowing in a given resistor for the corresponding values of potential difference  $V$  applied across the ends of resistor are given below in the table :

|               |     |     |     |     |     |
|---------------|-----|-----|-----|-----|-----|
| $I$ (Amperes) | 0.4 | 0.8 | 1.2 | 1.6 | 2.0 |
| $V$ (Volts)   | 1.2 | 2.4 | 3.6 | 4.8 | 6.0 |

Plot a graph between  $V$  and  $I$  and calculate the resistance of that resistor. 3

Source/editorial note: Visually confirmed complete replacement crop: question number, wording, all options/table values, mathematical symbols and printed marks. Whitespace normalized; mathematical glyphs represented in Unicode and stacked fractions as  $\frac{1}{4}$  and  $\frac{V}{4}$ . Page/QR/footer debris removed from text; marks retained in marks field. No supplied diagram; table retained in text and evidence. Answers not reviewed. Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

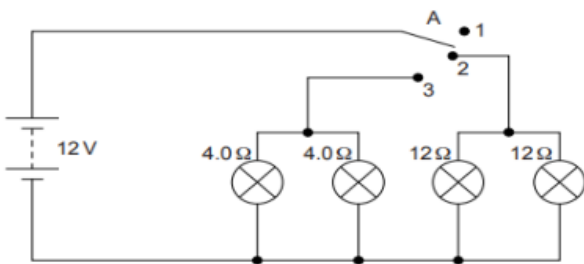
11. [3 marks] Source Q32 • 2024-25

|    |                                                                                                                                                                                                                                                                                       |   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 32 | <p>A. State the relationship between the resistance <math>R</math> of a wire to its length <math>l</math> and cross sectional area <math>A</math>. Use the mathematical symbols to arrive at the final formula.</p> <p>B. Using the formula define the resistivity of a material.</p> | 3 |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

Source/editorial note: The intermediate MS proportionality is printed  $R \propto l/A$ ; retained rather than silently changing it to  $R \propto 1/A$ . Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

Section D — 5-mark / case-based questions

12. [4 marks] Source Q39 OR iv • 2023-24

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 39 |  <p>Vinita and Ahmed demonstrated a circuit that operates the two headlights and the two sidelights of a car, in their school exhibition. Based on their demonstrated circuit, answer the following questions.</p> <p>(i) State what happens when switch A is connected to</p> <ol style="list-style-type: none"> <li>Position 2</li> <li>Position 3</li> </ol> <p>(ii) Find the potential difference across each lamp when lit.</p> <p style="text-align: center;"><b>OR</b></p> <p>(iv) Show, with calculations, which type of lamp, <math>4.0\ \Omega</math> or <math>12\ \Omega</math>, has the higher power.</p> | 1+1+2 |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|

Source/editorial note: Includes common (i), (ii) and OR (iv); 1+1+2 marks. Original scheme mislabels second line of (i) as (a). Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

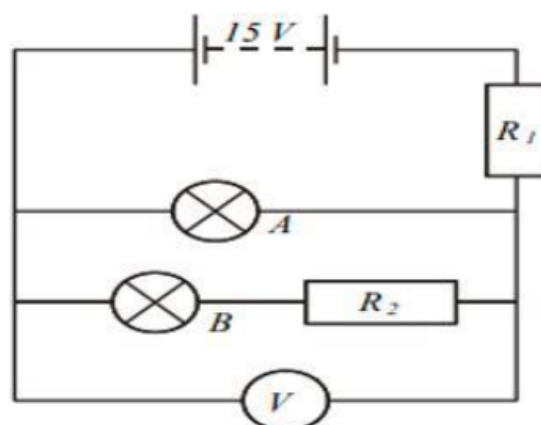
13. [5 marks] Source Q36 • compartment

36. (a) (i) Write the relation between the electrical resistivity ' $\rho$ ' and the resistance ' $R$ ' of a cylindrical conductor of length ' $l$ ' and area of cross-section ' $A$ '. Use this relationship to determine the SI unit of ' $\rho$ '.
- (ii) Find the electrical resistivity of the material of a wire of length 120 cm and diameter 0.4 mm. The resistance of this wire is  $3\ \Omega$ . 5

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

14. [5 marks] Source Q36 B • 2024-25

B.



As shown in the figure above A and B are two lamps. Lamp A is rated at 12 V, 24W. Lamp B is rated at 6.0 V. When lamp B operates at its rated voltage, the current in it is 3.0 A. The values of  $R_1$  and  $R_2$  are chosen so that both lamps operate at their rated voltages.

Based on the information given, answer the following.

- (i) Calculate the current in Lamp A.
- (ii) State and give reason for the reading of the Voltmeter.
- (iii) Calculate the resistance of  $R_2$ .
- (iv) Find the value of the resistance  $R_1$ .

Source/editorial note: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

# Solutions and verification status

Only explicitly verified solutions are printed. A missing solution is not a claim that the question is unsolvable.

## Question 1 — cbse\_sqp\_2019-20\_electricity\_q6

ii)  $40\ \Omega$ .  $V = IR$ ;  $V = 4\text{ V}$ ;  $I = 100\text{ mA} = 0.1\text{ A}$ .  $R = V/I = 4/0.1\ \Omega = 40\ \Omega$ .

|   |                                                                                                                                         |        |
|---|-----------------------------------------------------------------------------------------------------------------------------------------|--------|
| 6 | ii) = $40\ \Omega$ .<br>$V=IR$ , $V = 4\text{V}$ , $I = 100\text{ mA} = 0.1\text{ A}$<br>Hence $R = V/I = 4/0.1\ \Omega = 40\ \Omega$ . | 1 mark |
|---|-----------------------------------------------------------------------------------------------------------------------------------------|--------|

Marking-scheme PDF page(s): [1]

[https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2019\\_20/Science\\_MS.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_MS.pdf)

## Question 2 — sqp\_2022-23\_q13

(c) increases

|     |               |   |
|-----|---------------|---|
| 13. | (c) increases | 1 |
|-----|---------------|---|

Marking-scheme PDF page(s): [1]

[https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2022\\_23/Science-MS.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-MS.pdf)

## Question 3 — pyq-2024-a415e02467405f8b9fbacfc3

Official answer unavailable/withheld: No independently verified official answer supplied.

## Question 4 — SQP\_2026-27\_q31

C.  $4.0\text{ A}$

|    |                   |   |
|----|-------------------|---|
| 31 | C. $4.0\text{ A}$ | 1 |
|----|-------------------|---|

Marking-scheme PDF page(s): [7]

[https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2026\\_27/Science-MS.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf)

## Question 5 — pyq-2018-ac2b7b28762938a4a2df67a3

Official answer unavailable/withheld: No independently verified official answer supplied.

## Question 6 — cbse\_sqp\_2017-18\_electricity\_q26

Official answer unavailable/withheld: Official slope solution captured, but required plotted graph absent from official MS; not a complete publishable model answer.

Original marking scheme (see limitation above): [https://cbseacademic.nic.in/web\\_material/SQP/CLASS\\_X\\_2017\\_18/Science\\_MS.pdf](https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_MS.pdf)

## Question 7 — SQP\_2025-26\_q34\_B

B.

(i)  $1/R_P = 1/R_1 + 1/R_2 = 1/24 + 1/24 = 2/24$

$R_P = 12\text{ ohms}$

$R_T = R_P + 12 = 24\text{ ohms}$

$I = V/R = 6/24 = 0.25\text{ A}$

(ii) Same readings of  $A_1$  and  $A_2$

B.

(i)

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{24} + \frac{1}{24} = \frac{2}{24}$$

$$R_p = 12 \text{ ohms}$$

$$R_T = R_p + 12 = 24 \text{ ohms}$$

$$I = \frac{V}{R} = \frac{6}{24} = 0.25 \text{ A}$$

(ii) Same readings of  $A_1$  and  $A_2$

Marking-scheme PDF page(s): [7]

[https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2025\\_26/Science-MS.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf)

### Question 8 — cbse\_sqp\_2018-19\_electricity\_q26

LC of ammeter =  $0.2/10 \text{ A} = 0.02 \text{ A}$ . LC of Voltmeter =  $1/10 = 0.1 \text{ V}$ . Current =  $15 \times 0.02 \text{ A} = 0.3 \text{ A}$ . Potential difference =  $21 \times 0.1 \text{ V} = 2.1 \text{ V}$ .

$$26. \text{ LC of ammeter} = \frac{0.2}{10} \text{ A} = 0.02 \text{ A}$$

2

$$\text{LC of Voltmeter} = \frac{1}{10} = 0.1 \text{ V}$$

$$\text{Current} = 15 \times 0.02 \text{ A} = 0.3 \text{ A}$$

$$\text{Potential difference} = 21 \times 0.1 \text{ V} = 2.1 \text{ V}$$

Marking-scheme PDF page(s): [7]

[https://cbseacademic.nic.in/web\\_material/SQP/CLASS\\_X\\_2018\\_19/X\\_Science\\_MS\\_2018-19.pdf](https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_MS_2018-19.pdf)

### Question 9 — pyq-2025-74354640f102dee3b9237ff5

Official answer unavailable/withheld: No independently verified official answer supplied.

### Question 10 — pyq-2026-c2c822cabb523e97bd204b9e

Official answer unavailable/withheld: No independently verified official answer supplied.

Original marking scheme (see limitation above): [https://www.cbse.gov.in/cbsenew/marketing-scheme/2026/X/086\\_SCIENCE.zip](https://www.cbse.gov.in/cbsenew/marketing-scheme/2026/X/086_SCIENCE.zip)

### Question 11 — SQP\_2024-25\_q32

A.  $R \propto l$

$R \propto l/A$

$R = \rho l/A$ . (1.5)

B.  $R = \rho l/A$

$\rho = R A/l$  for  $A = 1 \text{ m}^2$  and  $l = 1 \text{ m}$ , we have

$\rho = R$

Hence resistivity is the resistance offered by a wire of length 1 m having a cross sectional area of  $1 \text{ m}^2$ . (1.5)

|    |                                                                                                                                                                                                                                                                                                                                                           |   |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 32 | <p>A.</p> $R \propto l$ $R \propto \frac{l}{A}$ $R = \rho \frac{l}{A} \quad (1.5)$ <p>B.</p> $R = \rho \frac{l}{A}$ $\rho = R \frac{A}{l} \text{ for } A = 1\text{m}^2 \text{ and } l = 1\text{m, we have}$ $\rho = R$ <p>Hence resistivity is the resistance offered by a wire of length 1 m having a cross sectional area of 1 m<sup>2</sup>. (1.5)</p> | 3 |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

Marking-scheme PDF page(s): [5]

[https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2024\\_25/Science-MS.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf)

### Question 12 — SQP\_2023-24\_q39\_OR\_iv

(i) 12  $\Omega$  lamps (only) on.

(a) 4  $\Omega$  lamps (only) on [0.5+0.5]

(ii) 12 V for both sets of lamps and all of them are in parallel. [1]

OR

$$P = V^2/R$$

All lamps are in parallel and hence same V for all lamps.

For 4  $\Omega$  lamps  $\rightarrow P = (12 \times 12)/4 = 36 \text{ W}$

For 12  $\Omega$  lamps  $\rightarrow P = (12 \times 12)/12 = 12 \text{ W}$

Hence 4  $\Omega$  lamps will have higher power. [0.5 x 4]

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 39 | <p>(i) 12 <math>\Omega</math> lamps (only) on.</p> <p>(a) 4 <math>\Omega</math> lamps (only) on [0.5+0.5]</p> <p>(ii) 12 V for both sets of lamps and all of them are in parallel. [1]</p> <p style="text-align: center;"><b>OR</b></p> <p><math>P = V^2 / R</math></p> <p>All lamps are in parallel and hence same V for all lamps.</p> <p>For 4 <math>\Omega</math> lamps <math>\rightarrow P = \frac{12 \times 12}{4} = 36 \text{ W}</math></p> <p>For 12 <math>\Omega</math> lamps <math>\rightarrow P = \frac{12 \times 12}{12} = 12 \text{ W}</math></p> <p>Hence 4 <math>\Omega</math> lamps will have higher power. [0.5 x 4]</p> | 4 |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

Marking-scheme PDF page(s): [6]

[https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2023\\_24/Science-MS.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-MS.pdf)

### Question 13 — pyq-2024-17a293a3889e84f2b1aa8c67

Official answer unavailable/withheld: No independently verified official answer supplied.

## Question 14 — SQP\_2024-25\_q36\_B

B.

(i) Use of  $P = IV$  (0.5)

$$I = P \div V = 24 \text{ W} \div 12 \text{ V}$$

Current in lamp A = 2 A (0.5)

(ii) Voltmeter reading = 12 V (0.5)

Lamp A and Lamp B are in parallel.

Hence p.d. across the arm containing A = p.d. across arm containing B

= 12 V (from a) (0.5)

(iii) p.d. across  $R_2$  + p.d. across B = 12 V. (0.5)

p.d. across B = 6 V (given)

Hence p.d. across  $R_2$  = 12 V - 6 V = 6V (0.5)

Current through  $R_2$  = Current through B = 3A (given) (0.5)

Use of  $R = V/I$

$$R_2 = 6\text{V}/3\text{A} = 2\Omega \text{ (0.5)}$$

(iv) Current through  $R_1$  = Total Current = 3A+2A = 5A

p.d. across  $R_1$  = 15V -12 V = 3V (0.5)

$$R_1 = 3\text{V}/5\text{A} = 0.6 \Omega \text{ (0.5)}$$

B.

(i) Use of  $P = IV$  (0.5)

$$I = P \div V = 24 \text{ W} \div 12 \text{ V}$$

Current in lamp A = 2 A (0.5)

(ii) Voltmeter reading = 12 V (0.5)

Lamp A and Lamp B are in parallel.

Hence p.d. across the arm containing A = p.d. across arm containing B

= 12 V (from a) (0.5)

(iii) p.d. across  $R_2$  + p.d. across B = 12 V. (0.5)

p.d. across B = 6 V (given)

Hence p.d. across  $R_2$  = 12 V - 6 V = 6V (0.5)

Current through  $R_2$  = Current through B = 3A (given) (0.5)

Use of  $R = V/I$

$$R_2 = 6\text{V}/3\text{A} = 2\Omega \text{ (0.5)}$$

(iv) Current through  $R_1$  = Total Current = 3A+2A = 5A

p.d. across  $R_1$  = 15V -12 V = 3V (0.5)

$$R_1 = 3\text{V}/5\text{A} = 0.6 \Omega \text{ (0.5)}$$

Marking-scheme PDF page(s): [9]

[https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2024\\_25/Science-MS.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf)

# Per-question source appendix

Original CBSE questions/images remain attributed to their source. Compilation selection and numbering are editorial. The full Science source PDFs may include unrelated chapters; only selected Electricity crops form this worksheet.

## 1. cbse\_sqp\_2019-20\_electricity\_q6

Class-X Science-086 SAMPLE QUESTION PAPER 2019-20 | SQP | Year/session: 2019-20 / 2019-20 | Paper code: Not recorded | Source question: 6 | Original section: SECTION A | Selected marks: 1 | PDF pages (one-based): 4

[https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2019\\_20/Science\\_SQP.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_SQP.pdf)

Also used in: official-01, official-06, official-07, official-10

## 2. sqp\_2022-23\_q13

CBSE Class X Science (086) Sample Question Paper 2022-23 | SQP | Year/session: 2022-23 / 2022-23 | Paper code: Not recorded | Source question: 13 | Original section: Section A | Selected marks: 1 | PDF pages (one-based): 4

[https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2022\\_23/Science-SQP.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-SQP.pdf)

Also used in: official-03, official-05, official-06, official-09, official-13

## 3. pyq-2024-a415e02467405f8b9fbacfc3

CBSE — Class X Science — 2024 — main — 31/4/1 | PYQ | Year/session: 2024 / main | Paper code: 31/4/1 | Source question: 14 | Original section: SECTION A | Selected marks: 1 | PDF pages (one-based): 7

<https://www.cbse.gov.in/cbsenew/question-paper/2024/X/SCIENCE.zip>

Also used in: official-02, official-06, official-07, official-09, official-12

## 4. SQP\_2026-27\_q31

CBSE Class X Science (086) Sample Question Paper 2026-27 (English) | SQP | Year/session: 2026-27 / 2026-27 | Paper code: Not recorded | Source question: 31 | Original section: Section - C | Selected marks: 1 | PDF pages (one-based): 11

[https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2026\\_27/Science-SQP.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-SQP.pdf)

Also used in: official-01, official-05, official-08, official-11, official-12

## 5. pyq-2018-ac2b7b28762938a4a2df67a3

CBSE — Class X Science — 2018 — 31/1 — main | PYQ | Year/session: 2018 / main | Paper code: 31/1 | Source question: 27 | Original section: Section - B | Selected marks: 2 | PDF pages (one-based): 12

[https://www.cbse.gov.in/curric~1/qpms2018/qp\\_10/Science.zip](https://www.cbse.gov.in/curric~1/qpms2018/qp_10/Science.zip)

Also used in: official-03, official-09, official-13

## 6. cbse\_sqp\_2017-18\_electricity\_q26

Sample Question Paper 2017-18 — Science — Class - X | SQP | Year/session: 2017-18 / 2017-18 | Paper code: Not recorded | Source question: 26 | Original section: SECTION - B | Selected marks: 2 | PDF pages (one-based): 3, 4

[https://cbseacademic.nic.in/web\\_material/SQP/CLASS\\_X\\_2017\\_18/Science\\_SQP.pdf](https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_SQP.pdf)

Also used in: official-04, official-10

## 7. SQP\_2025-26\_q34\_B

CBSE Class X Science (086) Sample Question Paper 2025-26 (English) | SQP | Year/session: 2025-26 / 2025-26 | Paper code: Not recorded | Source question: 34 B | Original section: Section - C | Selected marks: 2 | PDF pages (one-based): 12

[https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2025\\_26/Science-SQP.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf)

Also used in: official-02, official-07

## 8. cbse\_sqp\_2018-19\_electricity\_q26

Class-X Science-086 SAMPLE QUESTION PAPER-19 | SQP | Year/session: 2018-19 / 2018-19 | Paper code: Not recorded | Source question: 26 | Original section: SECTION E | Selected marks: 2 | PDF pages (one-based): 5

[https://cbseacademic.nic.in/web\\_material/SQP/CLASS\\_X\\_2018\\_19/X\\_Science\\_SQP\\_2018-19.pdf](https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_SQP_2018-19.pdf)

Also used in: official-02, official-10

## 9. pyq-2025-74354640f102dee3b9237ff5

CBSE — Class X Science — 2025 — main — 31/1/2 | PYQ | Year/session: 2025 / main | Paper code: 31/1/2 | Source question: 27 | Original section: SECTION C | Selected marks: 3 | PDF pages (one-based): 15

[https://www.cbse.gov.in/cbsenew/question-paper/2025/X/086\\_Science.zip](https://www.cbse.gov.in/cbsenew/question-paper/2025/X/086_Science.zip)

Also used in: official-05, official-08

## 10. pyq-2026-c2c822cabb523e97bd204b9e

CBSE Class X Science 2026 First Board 31-2-2 | PYQ | Year/session: 2026 / 2026 First Board | Paper code: 31/2/2 | Source question: 36 | Original section: Section C (Physics) | Selected marks: 3 | PDF pages (one-based): 29

<https://www.cbse.gov.in/cbsenew/question-paper/2026/X/Science.zip>

Also used in: official-03

## 11. SQP\_2024-25\_q32

CBSE Class X Science (086) Sample Question Paper 2024-25 (English) | SQP | Year/session: 2024-25 / 2024-25 | Paper code: Not recorded | Source question: 32 | Original section: Section-C | Selected marks: 3 | PDF pages (one-based): 9

[https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2024\\_25/Science-SQP.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf)

Also used in: official-06, official-11

## 12. SQP\_2023-24\_q39\_OR\_iv

CBSE Class X Science (086) Sample Question Paper 2023-24 (English) | SQP | Year/session: 2023-24 / 2023-24 | Paper code: Not recorded | Source question: 39 OR iv | Original section: Section-E | Selected marks: 4 | PDF pages (one-based): 8

[https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2023\\_24/Science-SQP.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-SQP.pdf)

Also used in: official-01, official-08, official-11, official-12

## 13. pyq-2024-17a293a3889e84f2b1aa8c67

CBSE — Class X Science — 2024 — compartment — 31/S/1 | PYQ | Year/session: 2024 / compartment | Paper code: 31/S/1 | Source question: 36 | Original section: SECTION D | Selected marks: 5 | PDF pages (one-based): 23

<https://www.cbse.gov.in/cbsenew/question-paper/2024-COMPTT/X/SCIENCE.zip>

Also used in: official-03, official-04, official-06, official-13

## 14. SQP\_2024-25\_q36\_B

CBSE Class X Science (086) Sample Question Paper 2024-25 (English) | SQP | Year/session: 2024-25 / 2024-25 | Paper code: Not recorded | Source question: 36 B | Original section: Section-D | Selected marks: 5 | PDF pages (one-based): 13

[https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2024\\_25/Science-SQP.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf)

Also used in: official-07